

Algebra 1
Unit 8
Lesson 3 Practice

Name **Answer Key** _____
Date _____
Period _____

Complete the table for each equation in 1-5.

Equation	Form	Key Feature(s) Revealed	Value(s) of Key Feature
1. $y = (x - 9)(x + 12)$	☐ Vertex form ☒ Factored form ☐ Standard form	y -intercept x -intercepts Direction graph opens	$(0, -108)$ $(9, 0)$ $(-12, 0)$ Graph opens up $A > 0$
2. $y = (x - 5)^2 - 8$	☒ Vertex form ☐ Factored form ☐ Standard form	y -intercept Vertex Axis of symm Direction of graph	$(0, 17)$ $(5, -8)$ $x = 5$ Opens up $A > 0$
3. $y = -\frac{1}{4}x^2 - 4x - 4$	☐ Vertex form ☐ Factored form ☒ Standard form	y -intercept Direction of graph	$(0, -4)$ Opens down $A < 0$
4. $y = 7(x - 1)^2$	☒ Vertex form ☐ Factored form ☐ Standard form	y -intercept Vertex Axis of symm Direction of graph	$(0, 7)$ $(1, 0)$ $x = 1$ Opens up $A > 0$
5. $y = -x^2 + 12x + 36$	☐ Vertex form ☐ Factored form ☒ Standard form	y -intercept Direction of graph	$(0, 36)$ Opens down $A < 0$

6. Rewrite the quadratic equation in factored form.

$$y = x^2 - 3x - 10$$

$$y = (x - 5)(x + 2)$$

7. Rewrite the quadratic equation in standard form.

$$\begin{aligned}y &= (x - 3)^2 - 7 \\y &= (x - 3)(x - 3) - 7 \\y &= x^2 - 6x + 9 - 7 \\y &= x^2 - 6x + 2\end{aligned}$$

8. Rewrite the equation in factored form.

$$\begin{aligned}y &= -x^2 + 11x - 28 \\y &= -(x^2 - 11x + 28) \\y &= -(x - 7)(x - 4)\end{aligned}$$

9. Rewrite the equation in standard form.

$$\begin{aligned}y &= -2(x - 1)(x + 9) \\y &= -2(x^2 + 8x - 9) \\y &= -2x^2 - 16x + 18\end{aligned}$$

10. Cyrus makes model rockets with his boy scout troop. The rockets launch straight up from 25 meters above the ground with a velocity of 20 meters/second. The height of the rocket is represented by the equation $y = -5x^2 + 20x + 25$, where x is time in seconds. Rewrite the equation in factored form.

$$\begin{aligned}y &= -5(x^2 - 4x - 5) \\y &= -5(x - 5)(x + 1)\end{aligned}$$